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Preserved Regeneration and Functional Recovery of the Injured Recurrent Laryngeal Nerve After Secondary Surgical Repair in Adult Rats

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Objectives: Transection of the recurrent laryngeal nerve (RLN) is accompanied by poor functional recovery, despite primary repair, because of regeneration difficulties. Nimodipine can promote regeneration, but it is not yet clear whether preoperative treatment is necessary. It is also not clear whether surgical repair following RLN injury may be performed in a second procedure, with preserved regeneration. This study investigated the time window for secondary surgical repair of the transected RLN and the need for preoperative administration of nimodipine.

Methods: In adult rats, the left RLN was transected and repaired at time intervals up to 3 weeks after transection, in combination with nimodipine treatment starting either before or after the operation. Regeneration and neuromuscular recovery were assessed by electrophysiology, retrograde tracing, and immunohistochemistry.

Results: Similar (whether 0, 2, or 7 days) regenerative results were obtained when the RLN was repaired up to 1 week after injury, given nimodipine administration, whereas fewer motor neurons managed to regenerate after nerve repair at 3 weeks after the initial transection. No beneficial effect was detected from preoperative nimodipine administration.

Conclusions: Provided that nimodipine is administered, surgical reconstruction of the RLN can be performed within 1 week after the initial nerve trauma, with preserved neuromuscular function. Nimodipine may be administered at the time of RLN transection injury.

Key Words: posterior cricoarytenoid muscle, recurrent laryngeal nerve, reinnervation.

INTRODUCTION

Transection of the recurrent laryngeal nerve (RLN) is accompanied by difficulties in recovering the neurophysiological and clinical function of the vocal folds, despite repair strategies.[1-3] Previous explanations have focused on factors such as neuronal death in the brain stem,[4-7] but experimental studies suggest that insufficient regeneration is an important parameter.[2,3,8,9] Poor reinnervation is most likely due to misguidance of axons to adductor and abductor intrinsic laryngeal muscles and/or slow sprouting and axonal elongation with resulting aberrant occupation of the denervated muscles, which has been shown previously.[10,11]

The timing of repair of an injured peripheral nerve has been the focus of several studies. After axotomy, the neuron changes its transcriptional machinery from a transmitting to a regenerative state, and begins to synthesize the proteins necessary for axonal elongation.[12] Axons start to sprout only hours after axotomy, but reach their maximum growth speed at a later point — a finding that has prompted investigations of the success behind conditioning lesions.[13-17] Furthermore, a secondary axotomy to an already injured nerve is not only required for optimizing the microsurgical alignment upon repair, but also triggers the neurons to a regenerative state.[18] Thus, there are data in favor of both primary and secondary repair after peripheral motor nerve injury, and the optimal time point for repair of the RLN is unknown.

Experimental and clinical data gathered over the past decade strongly suggest that the calcium channel inhibitor nimodipine accelerates axonal sprouting and improves neuromuscular function following peripheral motor nerve injury.[19-24] Recent data also show that nimodipine administration following experimental axonal injury to the RLN can improve regeneration and neuromuscular function[8] and can also improve the clinical outcome in patients subjected to complete RLN transection and repair.[25] Other data suggest that the serum concentrations

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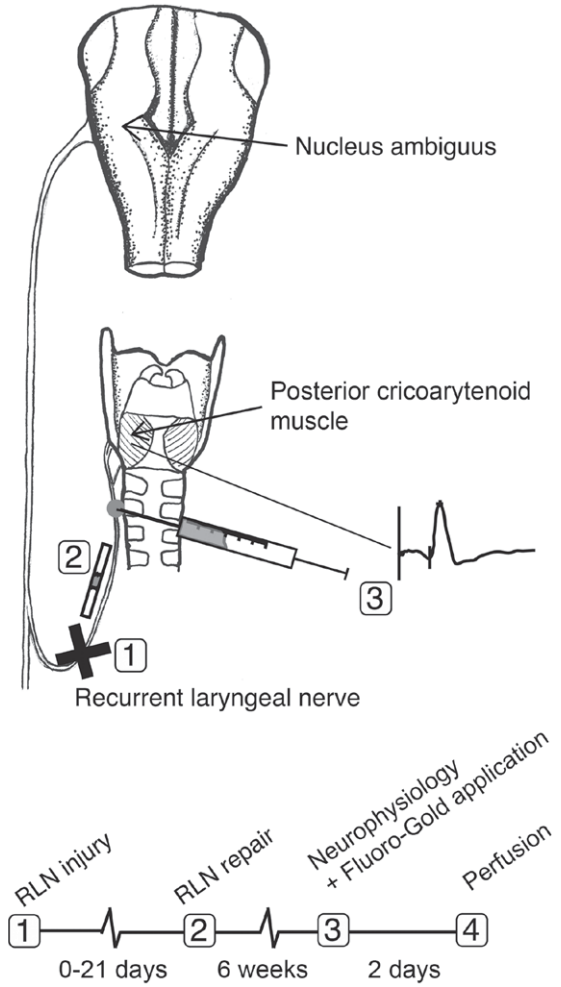


Fig 1. Schematic drawing of surgical methods. After initial recurrent laryngeal nerve (RLN) transection (1), RLN was grafted with autologous sural nerve graft at 0, 2, 7, or 21 days (2). Six weeks after repair, reinnervation was evaluated by electrophysiology, together with retrograde tracing with Fluoro-Gold (FG) in order to label central motor neurons that had regenerated through graft (3).

of nimodipine must be at a therapeutic level from the point of nerve injury if its full regeneration-promoting effect is to be realized,[23] but it is not known whether the administration of nimodipine must start before an RLN transection, or whether it may begin directly afterward.

The aims of the present study were to evaluate whether the effect of nimodipine administration differs depending on whether it begins before or after operation, and to investigate the time window for successful nerve repair following RLN transection. After nerve injury, repair was performed with grafting microsurgery at different time points following the initial lesion. Nimodipine treatment started either before or after the initial nerve lesion. Regeneration was monitored by retrograde tracing and functional assessment of brain stem neurons by immunohistochemistry, as well as electrophysiological evaluation of RLN function. The superior functional outcome resulting from nimodipine administration compared to no treatment following nerve injury has been studied elsewhere,[8,19-23,26] and was not in the scope of the present study.

MATERIALS AND METHODS

Forty-eight female Sprague-Dawley rats (250 to 300 g) were used. Nimodipine (Bayer Healthcare, Leverkusen, Germany) was administered continuously by mouth through rat pellets (1,000 ppm, Lactamin AB, Linkoping, Sweden), resulting in mean nimodipine serum concentrations of 6.3 ng/mL.[8] Nimodipine treatment started either 3 days before the initial nerve transection, or directly after. The rats were anesthetized by intraperitoneal injection of ketamine hydrochloride (Ketalar, Pfizer, New York, New York, 50 mg/kg body weight) and medetomidine (Domitor, Orion Animal Health, Stockholm, Sweden, 0.5 mg/kg body weight). The animals underwent microsurgery at 3 different time points (Fig 1).

SURGERY, ELECTROPHYSIOLOGY, AND RETROGRADE TRACING

RLN Injury. After a cervical ventral midline incision, the left RLN was identified, dissected free from surrounding tissue, and transected at the level of the seventh tracheal ring. The proximal stump was directed away from the tracheoesophageal groove and placed onto the ventral surface of the trachea.

RLN Repair. In order to collect an autologous nerve graft, we made an incision on the lateral side of the distal left hind limb and the sural nerve was identified, dissected free, and resected. Next, the proximal and distal endings of the transected left RLN were again exposed and carefully dissected free from surrounding tissue, placed in the tracheoesophageal groove, and subjected to axotomies followed by insertion of a 5-mm sural nerve graft segment to bridge the gap between the proximal and distal RLN. Care was taken to ensure full contact of the nerve endings. Fibrin sealant (Tisseel, Baxter, Deerfield, Illinois) was used to fixate the anastomosis.

Experimental Groups. The animals were randomized into the following groups: immediate repair (n = 12), repair after 2 days (n = 12), and repair after 7 days (n = 12). Within these groups, half of the animals started nimodipine treatment 3 days before the nerve lesion, and half directly after the lesion. Because of the similar regenerative and functional results in the groups above (nerve repair up to 1 week after transection), 1 additional group (n = 6) was added with nerve repair at 3 weeks following the

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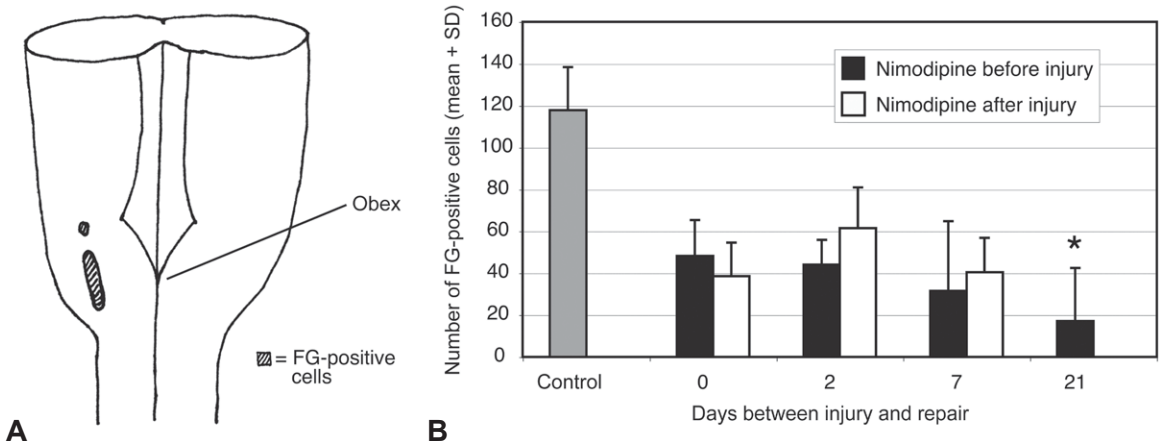


Fig 2. A) Schematic drawing of brain stem demonstrating location of FG-positive neurons in ipsilateral nucleus ambiguus, retrogradely traced from surgically repaired RLN. **B)** Number of FG-positive neurons (mean + SD) in nucleus ambiguus in brain stem, retrogradely traced from RLN, 6 weeks after nerve repair. When RLN was surgically repaired 3 weeks after initial trauma (asterisk), significantly fewer neurons regenerated through graft ($p < 0.05$, one-way analysis of variance) than in other groups.

initial lesion, combined with preoperative (3 days before nerve transection) nimodipine administration — that is, the theoretically most favorable nimodipine regimen. The reference group (n = 6) consisted of animals that had undergone neither injury nor treatment.

Electrophysiological Examination and Retrograde Tracing. Six weeks after the nerve repair, the animals were reanesthetized. The esophagus was separated from the trachea and larynx, and one of the intrinsic laryngeal muscles, the posterior cricoarytenoid muscle (PCA), was identified. A single steel hooked-wire electrode of 0.51 mm (Viasys Healthcare, Madison, Wisconsin) was inserted into the body of the left PCA in a rostral-to-caudal mode. The reference electrode was placed at the rostral end of the wound and connected to a Key Point workstation (Medtronic, Tolochenaz, Switzerland). The left RLN was stimulated at supramaximal intensity at a point proximal to the nerve graft, with a 1-mm needle electrode (0.7 mA; stimulation duration, 0.1 ms). The compound muscle action potential (CMAP) was recorded from the left PCA. After the electrophysiological examination, the RLN was transected distal to the nerve graft, close to the larynx, and the proximal stump was inserted into a polyethylene capsule containing 1 µL of 1% wt/vol Fluoro-Gold (FG; Fluorochrome Inc, Denver, Colorado). After 15 minutes, the capsule was removed and the area was rinsed with saline solution.

Perfusion. Two days after the tracing procedure, the animals were deeply anesthetized and painlessly sacrificed by intracardial perfusion of 200 mL body-warm saline solution followed by cold paraformaldehyde (4%) in phosphate buffer. The brain stem was removed and postfixed in the same fixative for 1.5 hours, then rinsed in saline solution and transferred to phosphate buffer (pH 7.4) containing 15% wt/vol sucrose, in which it was left overnight. The brain stems were transversely cut in sequences of 14 µm + 14 µm + 20 µm with a Leica CM3000 cryostat at –24°C, so that 5.1 mm was covered rostrocaudally (well enveloping the RLN area within the nucleus ambiguus).[9] The 20-µm sections were used for neuron quantification analysis, representing 20 µm/(20 µm + 14 µm + 14 µm) = 42% of the total volume of the nucleus.

QUANTIFICATION OF RETROGRADELY TRACED NEURONS

The FG was visualized with a Leica DM 4000B fluorescence microscope (lenses 20× and 40×). The tracer was seen assembled in the cytoplasm. Positively stained cells were counted manually in all serially cut 20-µm sections. To avoid the error of split cell bodies, we counted only those neurons with a clearly visible nucleus. Shrinkage of the brain stem due to neuronal loss may result in counting errors in serial sections; however, as previous results have shown that a similar injury (RLN resection) in the same system causes no neuronal loss,[9] we did not make any adjustments for this.

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The 14-µm sections of the brain stem were incubated with 1% bovine serum albumin, 0.1% sodium azide, and 0.3% Triton X-100 (Sigma Aldrich, Stockholm, Sweden) in phosphate buffer for 60 minutes at room temperature. The sections were then incubated overnight at 4°C in phosphate buffer containing 0.3% Triton X-100 with the following primary antibodies: 1) goat–anti–choline acetyltransferase (ChAT) 1:500 (Chemicon International,

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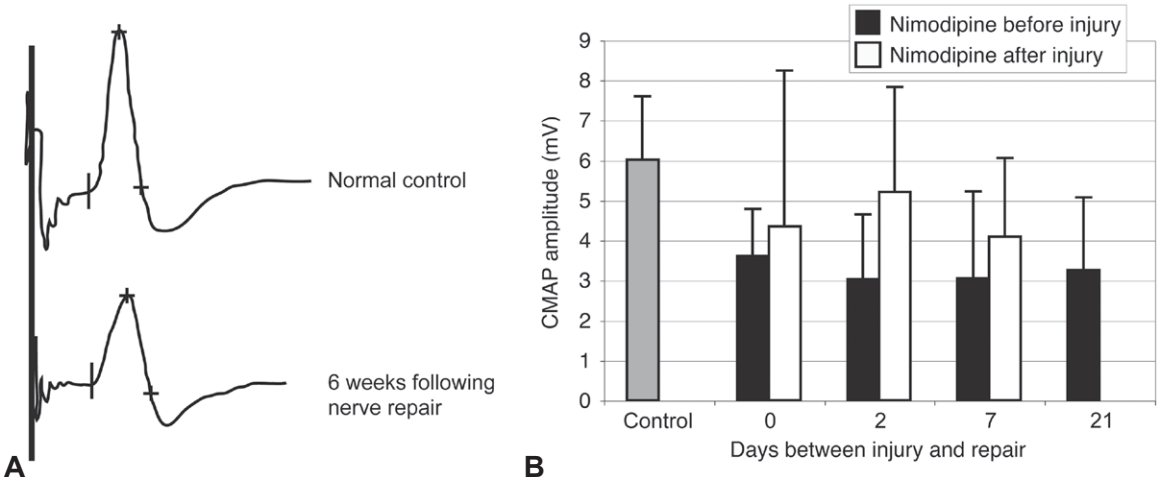


Fig 3. A) Representative compound muscle action potentials (CMAPs) recorded from posterior cricoarytenoid muscle after stimulation of RLN, in normal controls and 6 weeks after nerve repair. **B)** Amplitude of CMAPs (mean + SD) demonstrate large variations within groups subjected to RLN injury and repair. No differences could be detected between groups (one-way analysis of variance).

Temecula, California), and 2) goat–anti-synaptophysin 1:10,000 (gift from Dr R. Jahn, Max Planck Institute, Leipzig, Germany). To visualize immunoreactivity, we processed the sections with Cy3-conjugated secondary antibody (Jackson, West Grove, Pennsylvania) for 1 hour at room temperature. The sections were mounted in a 1:1 solution of glycerol and phosphate buffer.

RESULTS

RETROGRADE TRACING AND NEURON QUANTIFICATIONS

The FG-positive cells represented neurons projecting from the brain stem through the RLN to a point just before the entrance of the laryngeal muscles (ie, axons that had regenerated through the graft). In normal control animals, neurons containing FG were found in the nucleus ambiguus (NA), located 1.5 mm caudal to 1.8 mm rostral to the obex (the fusion point of the central canal into the fourth ventricle; Fig 2A). The neurons were distributed rostrocaudally in 2 groups, in line with earlier observations of RLN neurons.[9,27]

In animals subjected to RLN injury and repair, the number of neurons retrogradely traced with FG almost reached 50% of that seen in normal controls when they were surgically repaired up to 7 days after the initial lesion (Fig 2B), with preserved location within the NA. No statistical differences could be detected between the groups repaired within 1 week (one-way analysis of variance). No differences were observed between groups that received nimodipine before or after nerve transection. In the group in which the nerve was repaired 21 days after transection, fewer neurons were found in the NA than in the group with repair 2 days after transection ($p < 0.05$, one-way analysis of variance; Fig 2B).

ELECTROPHYSIOLOGY

In normal control animals, stimulation of the RLN resulted in reproducible CMAPs recorded from the PCA (Fig 3A). The amplitude of the CMAPs, reflecting the amount of muscle fibers that depolarize upon nerve stimulation and hence the neuromuscular function, was 6.01 ± 1.58 mV (mean ± SD; n = 6; Fig 2B).

At 6 weeks after nerve repair, all but 1 animal demonstrated reproducible CMAPs. The CMAPs were generally of a more broad-based appearance compared to normal controls, reflecting a population of nerve fibers of more varying conduction velocities (Fig 3A). The CMAP amplitudes varied considerably within groups, and no statistical differences could be detected between the groups (one-way analysis of variance), regardless of medical or surgical repair strategy (Fig 3B).

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Contralateral to the nerve injury, the RLN neurons were identified by size, shape, and topographic location within the NA. Ipsilateral to the nerve injury, the RLN neurons in the NA were identified by FG content in the cytoplasm (Fig 4A). Synaptophysin immunoreactivity (SP-IR), which represents areas of synaptic contact between neurons, was found throughout the brain stem sections. Contralateral and ipsilateral to the injury, SP-IR appeared intense throughout the neuropil in the NA, in close proximity to neuronal membranes on both the unoperated and the operated sides (Fig 4B,C). No difference in SP-IR could be observed between the unop-

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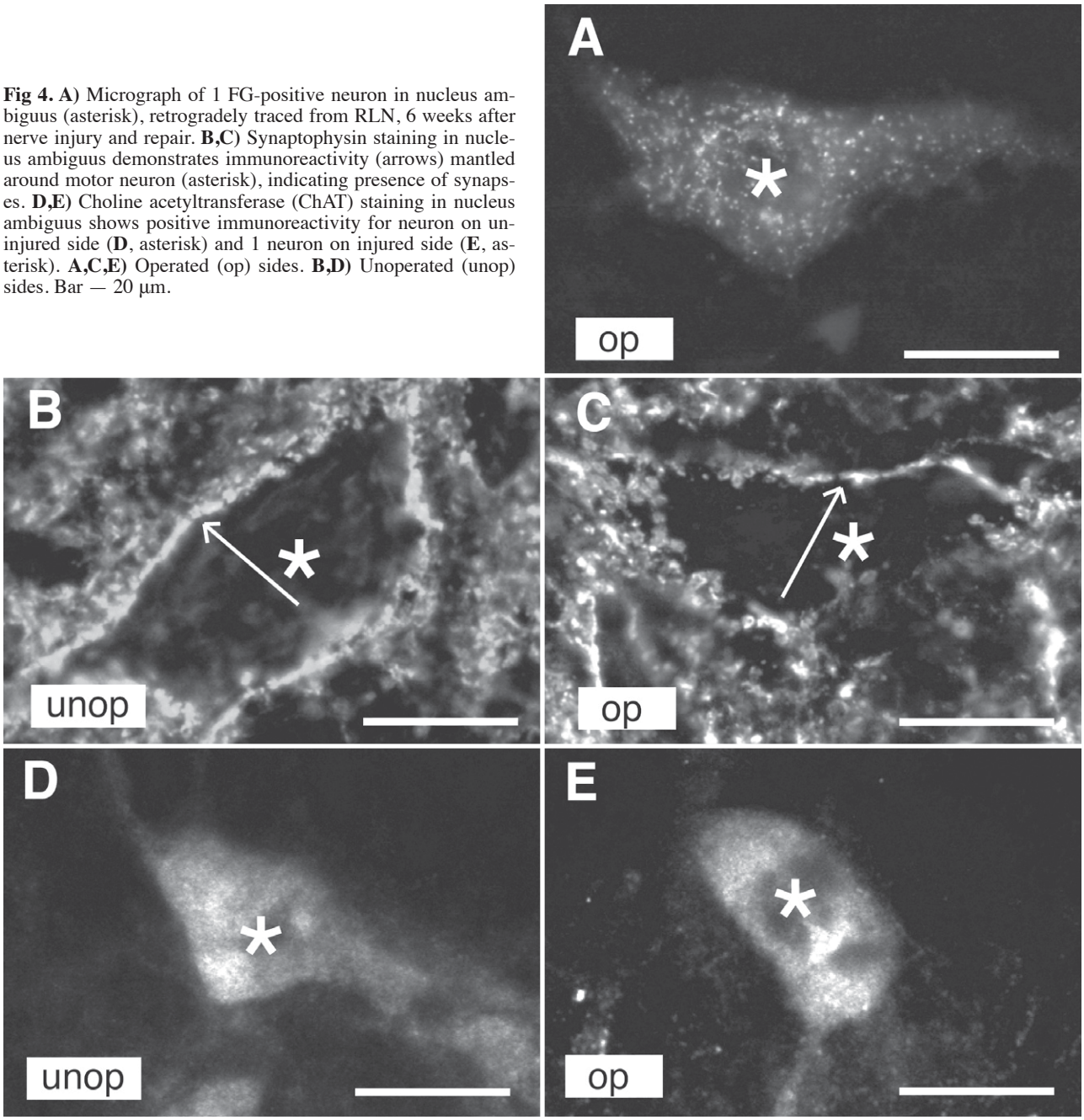


Fig 4. A) Micrograph of 1 FG-positive neuron in nucleus ambiguus (asterisk), retrogradely traced from RLN, 6 weeks after nerve injury and repair. **B,C)** Synaptophysin staining in nucleus ambiguus demonstrates immunoreactivity (arrows) mantled around motor neuron (asterisk), indicating presence of synapses. **D,E)** Choline acetyltransferase (ChAT) staining in nucleus ambiguus shows positive immunoreactivity for neuron on uninjured side (**D**, asterisk) and 1 neuron on injured side (**E**, asterisk). **A,C,E)** Operated (op) sides. **B,D)** Unoperated (unop) sides. Bar — 20 μm.

erated and operated sides at 6 weeks after surgical repair of the transected RLN, or in control animals.

Choline acetyltransferase immunoreactivity (ChAT-IR) was found in large multipolar motor neurons in the NA, dispersed throughout the cytoplasm, respecting the cell nucleus. Contralateral (Fig 4D), as well as ipsilateral (Fig 4E), to the nerve injury, cells with morphological characteristics of motor neurons in the NA appeared with positive ChAT-IR. On the operated side, all FG-containing neurons were positive for ChAT-IR (Fig 4E). In addition, ChAT-IR–positive neurons without FG content were found within the operated NA, probably representing motor neuronal input to a target other than the intrinsic laryngeal muscles. Negative control experiments (omission of primary or secondary antibodies) were negative for SP-IR and ChAT-IR.

DISCUSSION

This study investigated the regeneration and neuromuscular functional outcome following immediate or secondary surgical repair of the RLN, combined with treatment with the regeneration-promoting agent nimodipine, starting before or at the time of nerve injury. This experimental setting was designed to elucidate when to repair a damaged RLN

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and when to start nimodipine administration. The rationale was founded on existing strong evidence for the regeneration-promoting effect of nimodipine, which in this work was considered to be a pharmacologic basis for treatment of peripheral motor nerve injury.[19-26] The results demonstrate that the regeneration capacities and functional outcomes of the injured RLN are similar whether surgical repair takes place immediately or within 1 week after the initial lesion, and also that postoperative and preoperative administration of nimodipine are equally effective. Repair at 3 weeks after transection, however, resulted in poorer regeneration through the nerve graft, although this was not accompanied by a detectable difference in neuromuscular function.

The long-term functional outcome following a peripheral nerve injury depends on factors such as neural death, central synaptic reorganization, the regenerative capacity of the axons, the degree of misguided reinnervation, and the state of the target organ.[28-31] Surgical reconstructive treatment of peripheral nerve injuries aims at providing continuity of the epineural sheath in order to provide guidance for the outgrowing axons to their original targets.[29] In the case of RLN injury, which is not associated with a significant loss of central motor neurons,[9] poor functional restitution has been attributed to misguided or insufficient reinnervation and the subsequent loss of coordinated function of the intrinsic laryngeal muscles.[2,3]

There is evidence to suggest that the denervated PCA may be reinnervated by nerve fibers other than the original RLN. It is most likely that there is competition to reinnervate the denervated PCA. Existing evidence to support this theory includes the use of nimodipine to speed up regeneration within the injured RLN, which resulted in an increase of the RLN representation in the PCA (closer to the original neuroanatomy) and in improved neuromuscular function,[8] as well as electron microscopic evidence of aberrant reinnervation of the denervated PCA.[10,11]

Nimodipine is a voltage-gated calcium channel antagonist; its receptors are located on motor neurons and on their axons.[22] Calcium regulation in the growth cone is important for axonal elongation, and the regulation of calcium currents with calcium channel blockers has previously been shown to be useful in increasing the pace of regeneration, both in vitro and in vivo.[32,33] The regeneration-promoting effect of nimodipine has been demonstrated by several groups[8,19-23,25,26] and was not evaluated in the present study. However, there are data to suggest that the treatment must start before transection,[23] which is impractical in the clinical setting. This question was addressed by giving half of the animals nimodipine before transection, and the rest at the time of nerve injury.

Considering the time to reinnervation of the PCA following RLN injury as an important parameter, the current study challenges the use of secondary repair at a later time, provided that nimodipine is used. Because axons start to sprout within hours after an axonal injury,[28] immediate alignment and primary repair should be advantageous, but studies have also shown that the regenerative machinery within the neuron and axons is faster at a later time point,[34] and so secondary repair could also be favorable (see Introduction). The tracing studies revealed that about one third of the neuron pool in the NA regenerated through the graft and reached the PCA, regardless of whether the nerve was repaired immediately or at 2 or 7 days after transection (Fig 2B). Further, this one third of the neuron pool generated an electrophysiological response (CMAPs) of about 50% compared to control animals, also independent of the timing of surgery (Fig 3B). Because the same number of neurons projected through the graft, and these axons generated about the same electrophysiological response when surgically reconstructed up to 1 week after injury, we added a complementary group that underwent repair at 3 weeks after transection in combination with the theoretically most favorable nimodipine regimen (ie, preoperative start).[23] The results showed that a 3-week time period without RLN repair generated fewer motor neurons projecting through the graft, but with no obvious difference in electrophysiological response in the PCA, which would have been difficult to detect with the large variations within the groups.

The amplitude of the CMAPs varied considerably within groups (Fig 3B), being in some animals even higher than in normal controls (hyperinnervation). These results reflect individual differences in the number of nerve fibers that reached the PCA and the size of their motor units, possibly due to the complex neuroanatomy of the larynx, which allows axons from both the superior and inferior laryngeal nerves to supply the laryngeal muscles.[35]

A prolonged denervation time (months) is a known negative factor for the functional outcome.[36] The decline in reinnervation success was in this study detected at 3 weeks after the initial transection, but not during the first postoperative week. From a clinical perspective, the current data suggest that provided nimodipine is administered, similar (whether 0, 2, or 7 days) regeneration and functional recovery may be seen after RLN transection if the nerve is repaired during the first week, whereas repair thereaf-

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ter seems to hamper reinnervation.

CONCLUSIONS

The data from this study indicate that surgical reconstruction of the injured RLN can be performed within 1 week after the initial transection, with preserved regeneration and functional recovery of the RLN through an autologous nerve graft, provided that the regeneration-promoting agent nimodipine is administered. Nerve repair 3 weeks after RLN transection seems to result in worse regeneration. The current data cannot support previous observations that a preoperative start of nimodipine is superior to a postoperative start.

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